Matthew Hutchinson

Los Angeles | (415) 596-0301 | mahutchinson@ucla.edu | LinkedIn | Portfolio

Mechanical Engineer pursuing an M.S. at UCLA (Design, Robotics & Manufacturing), focused on physical AI: combining robot learning with robust hardware, sensing, and control. Hands-on experience with autonomy stacks, impedance control, and embedded systems.

EDUCATION

UCLA | Master of Science in Mechanical Engineering (DRoM) September 2025 – Present

Reinforcement Learning, Deep Learning, Optimal Control, Soft Robotics, Bionic Systems, Compliant Mechanisms, Data Science

UCLA | Bachelor of Science in Mechanical Engineering September 2022 – June 2024

Robotics (Kinematics, Dynamics & Control), Rapid Prototyping, Mechanism Design, Data Structures & Algorithms, C++, Python

EXPERIENCE

UCLA DevX September 2022 – June 2024

Autonomous Rover Developer

- Owned autonomy and behavior stack for a competition rover: built line-following logic, lost-line recovery, and intersection handling within a ROS2-based perception + control system (Raspberry Pi + Arduino)
- Implemented 30 Hz wheel-velocity PID on Arduino; tuned gains using step-response analysis and validated closed-loop performance under varying load and battery conditions
- Sized drive motors from first principles: computed ramp torque, applied 1.5× safety margin; rover climbed 15° ramp carrying 3 disks and completed full obstacle course in under 5 minutes
- Integrated electronics subsystem: spec'd battery pack, wired buck converter and 10A fuse; mitigated camera latency by reducing resolution and implementing adaptive velocity scaling

Robotic Arm for Automated Fiber Placement March 2024 – June 2024

Researcher

- Designed and implemented impedance controller in PyDrake for roller-on-surface contact task; achieved stable force regulation at 50 ± 10 N with damped disturbance rejection
- Diagnosed near-singular arm configuration caused by Bezier-surface trajectory; resolved by simplifying to planar pass to isolate and validate controller behavior
- Tuned stiffness and damping gains iteratively using end-effector position-error and contact-force plots; validated against bench push tests

UCLA Engineering MakerSpace April 2024 – July 2024

Digital Fabrication Artist

- Mentored 30+ students through design thinking and hands-on prototyping (laser cutters, 3D printers, CNC tools); led workshops on Illustrator and Fusion 360 and authored quick-reference guides that reduced equipment downtime

NASA Langley Research Center August 2022

Intern, AAM Safety Team

- Investigated safety and airspace integration challenges for Advanced Air Mobility and eVTOL aircraft; proposed a conceptual detect-and-avoid (DAA) architecture using neural networks and multi-sensor fusion, delivering a safety assurance framework to NASA engineers

SKILLS

Languages & Tools: Python, MATLAB, C++, PyTorch, GCP Compute Engine, PyDrake, Git, Arduino, SolidWorks

Robotics & Systems: PID control, impedance control, embedded systems (Raspberry Pi, Arduino), ROS2 (collaborative), digital fabrication (laser cutting, 3D printing)